

Experiment - 2

Data Wrangling II

Create an “Academic performance” dataset of students and perform the following operations using Python.

1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.
3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [2]: import pandas as pd
```

```
In [3]: df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\2\AcademicPerformance.csv")
```

```
In [4]: print(df)
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	
...	...	...	...	...	
2235	NaN	NaN	NaN	NaN	
2236	NaN	NaN	NaN	NaN	
2237	NaN	NaN	NaN	NaN	
2238	NaN	NaN	NaN	NaN	
2239	NaN	NaN	NaN	NaN	

	test preparation course	Year_Birth	math score	reading score	\
0	none	1970.0	72.0	72	
1	completed	1961.0	NaN	na	
2	none	1958.0	90.0	95	
3	none	1967.0	NaN	NaN	
4	none	1989.0	76.0	78	
...	...	...	...	...	
2235	NaN	NaN	NaN	NaN	
2236	NaN	NaN	NaN	NaN	
2237	NaN	NaN	NaN	NaN	
2238	NaN	NaN	NaN	NaN	
2239	NaN	NaN	NaN	NaN	

	writing score	Dt_Admission	College_Fees
0	74	6/16/14	\$84,835.00
1	A	6/15/14	\$57,091.00
2	93	5/13/14	\$67,267.00
3	44	05-11-2014	\$32,474.00
4	75	04-08-2014	\$21,474.00
...	...	...	...
2235	NaN	NaN	NaN
2236	NaN	NaN	NaN
2237	NaN	NaN	NaN
2238	NaN	NaN	NaN
2239	NaN	NaN	NaN

[2240 rows x 11 columns]

```
In [5]: print(df['math score'])
```

0	72.0
1	NaN
2	90.0
3	NaN
4	76.0
...	
2235	NaN
2236	NaN
2237	NaN
2238	NaN
2239	NaN

Name: math score, Length: 2240, dtype: float64

```
In [6]: print(df['math score'].isnull())
```

```
0      False
1       True
2      False
3       True
4      False
...
2235    True
2236    True
2237    True
2238    True
2239    True
Name: math score, Length: 2240, dtype: bool
```

```
In [7]: print(df['reading score'])
```

```
0      72
1      na
2      95
3     NaN
4      78
...
2235  NaN
2236  NaN
2237  NaN
2238  NaN
2239  NaN
Name: reading score, Length: 2240, dtype: object
```

```
In [8]: print(df['reading score'].isnull())
```

```
0      False
1      False
2      False
3       True
4      False
...
2235    True
2236    True
2237    True
2238    True
2239    True
Name: reading score, Length: 2240, dtype: bool
```

```
In [9]: missing_values = ["n/a", "na", "--"]
df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\2\AcademicPerformance.csv", na_v
```

```
In [10]: print(df['reading score'])
```

```
0      72.0
1      NaN
2      95.0
3      NaN
4      78.0
...
2235    NaN
2236    NaN
2237    NaN
2238    NaN
2239    NaN
Name: reading score, Length: 2240, dtype: float64
```

```
In [11]: print(df['reading score'].isnull())
```

```
0      False
1       True
2      False
3       True
4      False
...
2235     True
2236     True
2237     True
2238     True
2239     True
Name: reading score, Length: 2240, dtype: bool
```

```
In [12]: dataset = [11,41,20,3,101,55,68,97,99,6]
```

```
In [13]: sorted(dataset)
```

```
Out[13]: [3, 6, 11, 20, 41, 55, 68, 97, 99, 101]
```

```
In [14]: quantile1, quantile3 = np.percentile(dataset, [25,75])
```

```
In [15]: print(quantile1, quantile3)
```

```
13.25 89.75
```

```
In [16]: iqr_value = (quantile3 - quantile1)
```

```
In [17]: print(iqr_value)
```

```
76.5
```

```
In [18]: lower_bound_value = quantile1 - (1.5*iqr_value)
```

```
In [19]: upper_bound_value = quantile3 + (1.5*iqr_value)
```

```
In [20]: print(lower_bound_value, upper_bound_value)
```

```
-101.5 204.5
```

```
In [21]: from datetime import date
df['age'] = date.today().year - df['Year_Birth']
```

```
In [22]: df['Year'] = pd.DatetimeIndex(df['Dt_Admission']).year
df['E_L'] = date.today().year - df['Year']
```

```
In [23]: df.head(5)
```

Out[23]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	Year_Birth	math score	reading score	writing score	Dt_Admission
0	female	group B	bachelor's degree	standard	none	1970.0	72.0	72.0	74	6/11/2010
1	female	group C	some college	standard	completed	1961.0	NaN	NaN	A	6/16/2010
2	female	group B	master's degree	standard	none	1958.0	90.0	95.0	93	5/16/2010
3	male	group A	associate's degree	free/reduced	none	1967.0	NaN	NaN	44	05-11-2010
4	male	group C	some college	standard	none	1989.0	76.0	78.0	75	04-08-2010

```
In [24]: df['Fees$'] = df['College_Fees'].str.replace(',','').str.replace('$','').str.replace(
df['Fees_M$'] = df['Fees$'].apply(lambda X:round(X/1000000))
```

```
In [25]: df.head(5)
```

Out[25]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	Year_Birth	math score	reading score	writing score	Dt_Admission
0	female	group B	bachelor's degree	standard	none	1970.0	72.0	72.0	74	6/11/2010
1	female	group C	some college	standard	completed	1961.0	NaN	NaN	A	6/16/2010
2	female	group B	master's degree	standard	none	1958.0	90.0	95.0	93	5/16/2010
3	male	group A	associate's degree	free/reduced	none	1967.0	NaN	NaN	44	05-11-2010
4	male	group C	some college	standard	none	1989.0	76.0	78.0	75	04-08-2010

```
In [ ]:
```